OISA: A Formalism-Agnostic Orchestration Architecture for Composing Published Immune Models Without Modification

Nathan Foulquier
LBAI — Laboratoire de Biologie des Lymphocytes Auto-Immuns, Inserm U1227, Université de Bretagne Occidentale
Centre de Données Cliniques, CHU de Brest
Brest, France
Email: nathan.foulquier@univ-brest.fr

***Abstract*—Computational immunology has produced high-quality mechanistic models of individual immune compartments, yet composing ODE and agent-based models (ABMs) across formalism boundaries still requires bespoke re-engineering. Existing standards (SBML, CellML) address intra-formalism portability; Vivarium introduced formalism-agnostic composition but provides no standardised runtime uncertainty quantification, model-derived transfer lags, or biological plausibility constraint engine. We propose OISA — the Orchestrated Immune Simulation Architecture — comprising (i) the ISSL, a formalism-agnostic JSON-LD checkpoint format with embedded runtime UQ (uncertainty bounds propagated at every tick via the `ci_95` schema field); (ii) a declarative configuration graph with support for model-derived transfer lags on edges; and (iii) an orchestrator engine with global clock synchronisation, causal DAG resolution, and biological plausibility enforcement.**

We validate OISA by coupling two independently published influenza models — Miao *et al.* 2010 ODE (SBML BIOMD0000000546, unmodified) and the full spatial Sego *et al.* 2020 ABM (CompuCell3D 4.8.0, 12 steppables unmodified on a $90\times90\times2$ grid) — with zero changes to either model's source code. Adapter layers total ~300 lines. The ABM runs as a rolling ensemble of $N = 20$ parallel CC3D instances (concurrent subprocesses, distinct MersenneTwister seeds); runtime uncertainty bounds — reported as the 2.5th–97.5th percentile across $N = 20$ stochastic CC3D replicates (2.5–97.5th percentile, see §V-A) — originate in the stochastic ABM and propagate through the ODE at each checkpoint. Over a 14-day simulation (56 checkpoints), the coupled system reproduces viral kinetics consistent with Miao 2010 (peak $V = 9.0\times10^6$ copies/mL at day 2.25, clearance to 0.05% of peak by day 13.75) and spatial immune recruitment (median $n_{\text{immune}} = 6$ [IQR: 5–7] at day 1, growing to 57 [IQR: 52–59] at day 13), with immune temporal lag and CTL-mediated clearance acceleration emerging from the bidirectional coupling. To test that this composition is structural rather than an artefact of the ODE–ABM pair, we further couple a third, categorically distinct formalism — a continuous-time stochastic Boolean signalling network (MaBoSS `TregModel`, 35 nodes) — through the same contract; each formalism reports its native uncertainty (ensemble percentiles for the ODE/ABM, a Wilson interval for the Boolean model) and these compose, verifiably, into the coupled output, giving what is to our knowledge the first heterogeneous cross-formalism runtime UQ in an immune multi-model simulation. This validation targets orchestration correctness — signal routing, causal ordering, and runtime UQ propagation. As a secondary check, emergent temporal features of the coupled system (immune-recruitment onset, viral-clearance day) show face-validity against independent published murine influenza kinetics; this is not biological prediction, as no parameter of the coupled configuration was fit to simultaneous in-vivo measurements.

***Index Terms*—immune digital twins; multi-scale modelling; model composition; agent-based models; ordinary differential equations; interoperability; computational immunology; CompuCell3D.**

I. Introduction

Immune-mediated diseases are inherently multi-compartmental phenomena. Influenza infection unfolds across scales: within a single tissue, viral replication dynamics interact with innate immune cytokine signalling and adaptive CTL recruitment, each operating on different timescales and representable in different formalisms. ODEs are natural for systemic viral kinetics and antibody dynamics, while ABMs are necessary where stochastic single-cell fate decisions dominate, as in tissue-level immune recruitment. The literature has consequently produced a collection of high-quality compartment-specific models that cannot communicate with one another.

Three bodies of prior work address adjacent problems. The COMBINE standards ecosystem (SBML [1], CellML [2], SED-ML [3]) provides portable representations for individual models but addresses intra-formalism composition only: models must be expressed in the same declarative format, ruling out ABMs written in Python or Julia. Vivarium [4] introduced formalism-agnostic composition with hierarchical stores and dynamic topology (discussed in §II-A), but provides no standardised runtime UQ propagation, model-derived transfer lags, or biological plausibility constraint engine. The CURE guidelines [5] define what properties a credible model should exhibit — Credibility, Understandability, Reproducibility, Extensibility — but prescribe no runtime infrastructure for heterogeneous composition (note: [5] is currently available as a preprint pending peer review). The cross-formalism, multi-timescale coordination problem remains open.

We propose OISA — the Orchestrated Immune Simulation Architecture — to fill this gap. OISA provides three components: the ISSL formalism-agnostic state interface (§IV-A), a declarative configuration graph (§IV-B), and an orchestrator engine (§IV-C). We demonstrate the architecture by coupling two independently published influenza models, using the full spatial Sego 2020 CompuCell3D ABM without any modification to its 12 biological steppables, and evaluate causal ordering, signal flow correctness, and biological plausibility in §V.

The contributions of this work are:

1) **The ISSL**, a formalism-agnostic checkpoint format with embedded provenance and runtime uncertainty quantification (ensemble bounds propagated at each tick from stochastic ABM through deterministic ODE via the `ci_95` schema field);
2) **A declarative configuration graph** with support for model-derived transfer lags on edges, where a lightweight transfer model computes the signal delay dynamically rather than using a fixed constant;
3) **A runtime orchestrator** with global clock synchronisation, causal DAG resolution, and biological plausibility constraint enforcement;
4) **Empirical demonstration** that two independently published models (Miao 2010 SBML ODE + Sego 2020 CC3D spatial ABM) can be composed with zero source modifications using ~300 lines of adapter code, with runtime UQ propagated end-to-end;
5) **Cross-formalism generalisation**: a third, categorically distinct formalism (a stochastic Boolean MaBoSS signalling network) coupled through the same contract, demonstrating heterogeneous cross-formalism uncertainty propagation (ensemble percentiles for the ODE/ABM and a native Wilson interval for the Boolean model) — to our knowledge the first such demonstration in immune multi-model simulation.

II. Related Work

A. Multi-Formalism Immune Simulation Frameworks

Several frameworks have addressed multi-scale composition in computational biology. Vivarium [4] introduced a port-based, formalism-agnostic composition interface with hierarchical stores, dynamic topology changes at runtime, and a mature Python ecosystem. OISA shares Vivarium's design principle of formalism-agnostic composition but addresses three capabilities not present in the Vivarium architecture: (i) a standardised inter-model signal format (ISSL) embedding runtime uncertainty quantification (ensemble bounds propagated at each checkpoint from a rolling ABM ensemble through the ODE via the `ci_95` schema field), (ii) model-derived transfer lags on edges (where a lightweight transfer model computes the signal delay dynamically rather than using a fixed constant), and (iii) a biological plausibility constraint engine operating at the composition level (mass conservation, parameter bounds, and divergence monitoring across coupled models). Vivarium's store-based data sharing and dynamic topology management are complementary capabilities not replicated by OISA.

PhysiCell [6] and PhysiBoSS [7] are the closest prior art to the three-formalism configuration reported here: PhysiBoSS embeds MaBoSS Boolean signalling networks [8] inside PhysiCell agents, coupling a continuous-time stochastic Boolean formalism to an agent-based one. The coupling, however, is *intra-engine*: the Boolean layer is compiled into the ABM and shares its address space, timestep, and build system, so the two formalisms cannot be developed, versioned, or uncertainty-quantified independently. OISA instead couples an *externally published* MaBoSS model (`TregModel`, distributed in the pyMaBoSS reference suite [9]) to the Miao2010 ODE and the Sego2020 ABM through the ISSL contract alone, each model running unmodified in its own engine. Because PhysiBoSS fuses the formalisms at compile time it exposes a single homogeneous stochastic-ABM uncertainty; OISA preserves and *propagates each formalism's native uncertainty definition across the coupling boundary* (§V-C). The rapid community-driven SARS-CoV-2 tissue simulator [10] — from which Sego *et al.* 2020 [11] derives — demonstrated that tissue-scale ABMs can be developed and shared rapidly, but required a uniform CompuCell3D substrate for all constituent models; coupling to external ODE models required bespoke adapter code outside the framework. Miao *et al.* 2010 [12] provided a rigorously calibrated ODE model of murine influenza kinetics embedded in the BioModels database (BIOMD0000000546); no standard mechanism exists to couple it to published tissue ABMs without rewriting either model. OISA demonstrates that such coupling is achievable by adding only a thin Emit/Accept interface layer to each model, with the full CC3D spatial ABM running unmodified as an independent process.

B. Simulation Interoperability Standards

The COMBINE standards ecosystem [13] has achieved significant progress on individual model portability. SBML [1] has accumulated over 1,000 curated models in BioModels; CellML [2] has analogous achievements in cardiac and physiological modelling; SED-ML [3] provides a standard for encoding simulation experiments. Table I compares these standards against OISA. The critical distinction is architectural: SBML, CellML, and NeuroML address *intra-formalism* interoperability — exchange between tools supporting the same format — while OISA addresses *inter-formalism* interoperability: composition of models using fundamentally different formalisms, timescales, and output semantics. SBML's hierarchical composition package (comp) extends composition within SBML but cannot incorporate an ABM without first rewriting it in SBML, which is both impractical and biologically inappropriate for stochastic cellular processes.

BioSimulators [14] and the SED-ML/COMBINE stack standardise *how a simulation experiment is described and re-*

executed across engines, and can drive heterogeneous solvers via a common execution API. Their scope is reproducible single-model (or intra-formalism co-simulation) execution; they do not define a runtime inter-model signal carrying an uncertainty interval, and they do not address the composition of a deterministic ODE, a stochastic ABM, and a stochastic Boolean model into one coupled run whose outputs carry uncertainty inherited from all three. The specific gap OISA fills is therefore neither model portability (SBML/CellML), nor experiment reproducibility (SED-ML/BioSimulators), nor port-based composition (Vivarium), but **heterogeneous cross-formalism uncertainty propagation**: a single transport contract under which each formalism reports its *native* notion of uncertainty and the orchestrator composes those intervals at coupling boundaries without homogenising them (§V-C, Table V).

C. Immune Digital Twins

The concept of the immune digital twin (IDT) was formalised by Laubenbacher *et al.* [15] as a continuously-recalibrated patient-specific computational model of the immune system. Subsequent reviews [16], [17] have identified interoperability as the primary technical barrier to clinical deployment, a conclusion echoed by the National Academies [18], by model integration consortia [19], and by multi-scale digital twin frameworks in adjacent domains [20]. OISA is the first runtime orchestration architecture designed specifically for heterogeneous immune model composition, providing the coordination infrastructure that these reviews identify as lacking.

III. Problem Formalisation

A. Formal Model Definition

We define a composable simulation model M as a triple $M = (S, \mathrm{Emit}, \mathrm{Accept})$, where S is the model's internal state space, $\mathrm{Emit}: S \times T \rightarrow \mathrm{ISSL}$ maps the current state and simulation time to a structured ISSL record, and $\mathrm{Accept}: \mathrm{ISSL} \rightarrow S$ updates the model state in response to an incoming inter-model signal. Any model — ODE, ABM, hybrid, or neural surrogate — that implements Emit and Accept is OISA-compliant without any other modification. Formalism-agnosticism follows directly: since the ISSL is the only communication channel, models need not know each other's internal representation, only the schema of the signal they receive. This design principle — letting each biological process be modelled in its natural formalism and solving the composition problem at the interface — avoids forcing biologically inappropriate representations on individual compartments.

B. The Composition Problem

A multi-model composition is a directed graph $G = (V, E)$ where each vertex v is a model M_v and each directed edge (u, v) specifies that a subset of u's `export_signals` are routed to v's Accept function, with optional transfer-model lags. The composition problem reduces to three coordination challenges:

1) **Heterogeneous time steps.** A global simulation clock (GSimT) must coordinate execution such that no model receives a signal purporting to come from a future GSimT tick. In the influenza reference implementation, the viral dynamics ODE steps at 6 h intervals while the tissue immune ABM steps at 24 h intervals (72 Monte Carlo Steps at 20 min/MCS); the GSimT tick is set to the GCD of all model Δt_i values (here 6 h).
2) **Stochastic–deterministic reconciliation.** Signals from a stochastic ABM to a deterministic ODE must carry uncertainty information. The orchestrator normalises all inter-model signals to (median, uncertainty bounds, unit) before routing, stored in the `ci_95` schema field: for stochastic models, bounds are computed from a rolling ensemble of N parallel instances (§V-A; with $N = 20$, reported as 2.5th–97.5th percentile); for deterministic models, bounds are derived by propagating the input uncertainty through the model equations.
3) **Causal ordering with feedback.** The immune model emits cytokine signals that drive viral clearance; viral load in turn drives immune recruitment — a feedback loop. The orchestrator resolves this by applying a one-tick delay on back-edges of the DAG, preventing circular dependency while preserving biological causality at the GSimT timescale.

IV. The OISA Architecture

A. The Internal Simulation State Log (ISSL)

The ISSL is a JSON-LD document emitted by a model at each checkpoint, structured in six sections: (i) `envelope` (model ID/version, GSimT timestamp, schema URI, formalism tag — validated and dispatched by the orchestrator before any payload is read); (ii) `continuous_state` (running entity populations with scaled `count`, `unit`, and ensemble `ci_95` bounds; `label` is model-local, mapped via signal-id routing); (iii) `discrete_events` (punctual events such as cell death or cytokine peaks, drawn from a controlled `event_type` vocabulary, with `n_realisations` for ABM events); (iv) `export_signals` (inter-compartmental signals available for routing: `signal_id` as routing key, `value` in declared `unit` — always biologically scaled regardless of formalism); (v) `internal_parameters` (kinetic parameter values with `provenance` DOI/URI to the calibration source); and (vi) `watchdog` (model health and `divergence_score`; values $> 0.15 \Rightarrow$ `PAUSE`). JSON-LD enables both human readability and machine-parsable semantic annotation via linked-data context, letting orchestrator components parse records without prior knowledge of any model's internal variable names.

Box 1 — ISSL excerpt, ensemble-aggregated Miao 2010 ODE checkpoint at day 1 (`count` = median, `ci_95` = 2.5th–97.5th percentile over $N = 20$ replicates):


```
{
 "envelope": {
   "model_id": "miao2010_ode",
   "model_version": "BIOMD0000000546",
   "sim_time_s": 108000,
```

TABLE I

COMPARISON OF SBML, CELLML, NEUROML/LEMS, AND OISA ACROSS CAPABILITIES RELEVANT TO MULTI-ORGAN IMMUNE SIMULATION. "—" = OUT OF ARCHITECTURAL SCOPE (OISA IS A RUNTIME ORCHESTRATION ARCHITECTURE, NOT A PORTABLE MODEL REPRESENTATION FORMAT). RUNTIME UQ: ENSEMBLE BOUNDS (REPORTED IN THE `ci_95` SCHEMA FIELD) ORIGINATE IN THE STOCHASTIC ABM ENSEMBLE ($N = 20$ PARALLEL CC3D INSTANCES; REPORTED AS 2.5–97.5TH PERCENTILE ESTIMATES) AND PROPAGATE THROUGH THE DETERMINISTIC ODE VIA TRIPLE INTEGRATION AT EACH CHECKPOINT (§V-A). "SBML L3" REFERS TO THE FORMAT SPECIFICATION ALONE; SED-ML [3] PROVIDES CO-SIMULATION CAPABILITIES INCLUDING HETEROGENEOUS TIME-STEPPING FOR SBML-COMPATIBLE MODELS BUT DOES NOT EXTEND TO ABM COMPOSITION.

Capability	**SBML L3**	**CellML 2.0**	**NeuroML/LEMS**	**OISA (ISSL)**
Portable model representation	✓	✓	✓	—
ODE / biochemical networks	✓	✓	neurons only	via `Emit()`
Agent-based models	×	×	×	via `Emit()`
Heterogeneous Δt between models	×	×	×	✓
Model-derived transfer lag on edges	×	×	×	✓
Inter-formalism composition (ODE + ABM)	×	×	×	✓
Runtime UQ propagation across models	×	×	×	✓
OBO ontology annotation	✓	✓	✓	✓ + wildcard
ABM scaling factor declaration	×	×	×	✓
Requires model rewrite to adopt	✓	✓	✓	× (add `Emit` only)


```
   "formalism": "ODE",
   "schema_uri": "schemas/issl_v1.schema.json"
 },
 "continuous_state": [
   {"label":"Ep", "count":545366.0, "unit":"cells",
    "ci_95":[545357.0, 545372.0]},
   {"label":"Eps","count":31064.6,  "unit":"cells",
    "ci_95":[31041.3, 31098.4]},
   {"label":"V",  "count":3.29e5,   "unit":"copies/mL",
    "ci_95":[3.285e5, 3.289e5]}
 ],
 "export_signals": [
   {"signal_id":"miao2010.viral_load",
    "value":3.29e5,"unit":"copies/mL",
    "ci_95":[3.285e5,3.289e5],"transfer_lag_s":null}
 ],
 "watchdog":{"status":"OK","divergence_score":0.0}
}
```


Note on `sim_time_s`: the ODE-local clock *after* the step completes; at the day 1 boundary (86,400 s) it reads 108,000 s, one 6 h ODE step beyond.

The ABM adapter emits the same ISSL envelope with `formalism="ABM"`, `model_version` pinned to `covid-tissue-models@5b7e42c+OISA_bridge`, a `grid` field (`"90x90x2"`), and export signals `sego2020.immune_cell_count` and `.recruitment_cytokine` (example day-1: median 6 immune agents [ensemble range 2–10]).

The SBML file `BIOMD0000000546_model1.xml` is loaded verbatim by libroadrunner; no XML modifications are made. The adapter calls `rr.simulate(t0,t1,2)` and reads species concentrations via the standard roadrunner API. For the Sego 2020 CC3D ABM, the adapter launches CompuCell3D 4.8.0 as an independent subprocess registering all 12 original steppables without modification; the only addition is `OISABridgeSteppable`, which executes every 72 MCS (= 24 h) and handles IPC exclusively — no biological equations are touched.

B. The Configuration Graph

The configuration graph is a YAML or JSON-LD file that fully specifies the composition — the orchestrator's sole input at initialisation; no model needs to know about any other model. It declares six elements: `models[]` ({id, formalism: ODE|ABM, executable, issl_port, delta_t_s}), `edges[]` ({source_model, signal_id, target_model, lag: `"constant:N"` | `"model:ID"`}), `transfer_models[]` ({id, formalism, executable, input_signal, output_signal, lag_output_field}), `global_clock` ({start_s, end_s, checkpoint_interval_s}), `calibration` ({data_source_uri, patient_id, recalibration_trigger, biomarker_map[ ]}), and `wildcard_namespace` ({prefix, registry_uri: null}, allowing non-OBO entities such as novel murine cytokines or patient-specific biomarkers). The declarative separation of identity, wiring, and temporal parameters implements the CURE extensibility requirement: any model node can be substituted without modifying the graph structure.

Influenza coupling graph (excerpt):


```yaml
models:
  - id: miao2010_ode
    formalism: ODE
    executable: models/ode_miao2010/miao2010_adapter.py
    delta_t_s: 21600          # 6 h
  - id: sego2020_abm
    formalism: ABM
    executable: models/abm_sego2020/sego2020_adapter.py
    delta_t_s: 86400          # 24 h  (72 MCS x 20 min/MCS)

edges:
  - source: miao2010_ode
    signal_id: miao2010.viral_load
    target: sego2020_abm
    lag: "constant:0"         # same-tick delivery
  - source: sego2020_abm
    signal_id: sego2020.immune_cell_count
    target: miao2010_ode
    lag: "constant:21600"     # one-tick delay (causal back
        -edge)

global_clock:
  start_s: 0
  end_s: 1209600            # 14 days
  checkpoint_interval_s: 21600
```


C. The Orchestrator Engine

The orchestrator is a nine-component server process that reads the configuration graph at startup, connects to all model processes, and manages the run. Components: (1) ISSL ingestion (JSON-LD + SI units + Mahalanobis OOD);

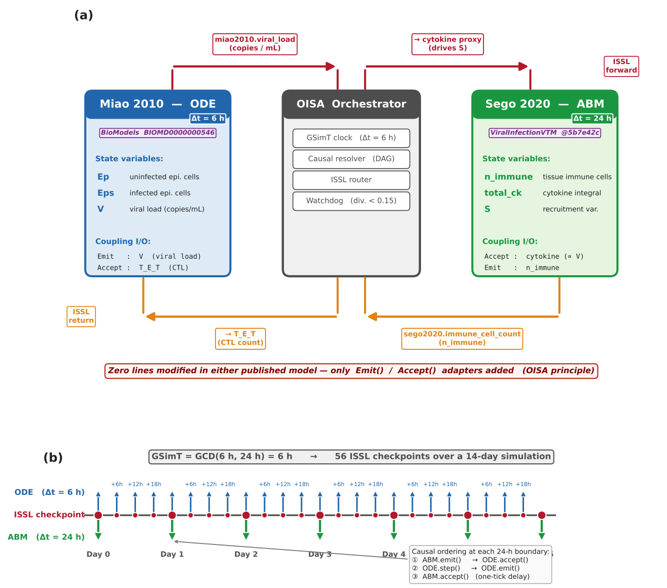

Fig. 1. OISA workflow for the influenza coupling reference implementation. **(a)** Architecture diagram: Miao 2010 ODE (SBML, BIOMD0000000546) and Sego 2020 CC3D ABM (CompuCell3D 4.8.0, GitHub @5b7e42c) connected through the OISA Orchestrator, with ISSL signal arrows annotated with `signal_id` and biological units. **(b)** GSimT timeline for a representative 5-day window: ODE ticks every 6 h, ABM ticks every 24 h = 72 MCS, ISSL checkpoints (orange dots), and causal-ordering annotation (`ABM.emit` → `ODE.accept` → `ODE.step` → `ODE.emit`) at the 24 h boundary.

(2) temporal scheduler (priority queue by `next_step_due`); (3) causal resolver (topological sort + one-tick back-edges); (4) constraint engine (mass conservation + parameter bounds); (5) state registry (versioned checkpoint snapshots + provenance); (6) transfer dispatcher (`SignalQueue`); (7) calibration bridge (EHR ingest, deferred); (8) output aggregator (GIS → checkpoint log + UQ propagation); (9) watchdog (health poll, pause/resume/rollback). Components 1, 4, 5, 9 have their core functionality implemented and covered by the published test suite (79 tests); sub-capabilities flagged as deferred (OOD detection, `ROLLBACK` recovery, PROV-O URIs, conservation-violation auto-rollback) are interface-specified but not yet in the test suite.

Temporal scheduling. At each GSimT tick (the GCD of all model Δt_i, here 6 h), the scheduler dispatches step commands to models whose `next_step_due` equals the current GSimT. The Sego 2020 CC3D ABM, with $\Delta t = 24$ h (72 MCS $\times$ 20 min/MCS), receives step commands every fourth tick. The ODE receives a step command at every tick.

Causal resolution. The causal resolver constructs a DAG from the configuration graph and performs topological sorting to determine execution order within each GSimT tick. At each 24 h boundary (every fourth tick), the execution order is: (1) `ABM.emit()` — export current immune cell count from CC3D cell inventory; (2) `ODE.accept()` — inject `n_immune` $\rightarrow T_{E_T}$ in SBML; (3) `ODE.step(`Δt`)`; (4) `ODE.emit()` — export viral load. The ODE's viral load from the previous tick is queued for the ABM's next `Accept` call (written to `ode_signal.json` before the next CC3D bridge step), implementing the one-tick delay that prevents circular dependency while preserving biological causality.

Calibration bridge (component 7). The calibration bridge has its interface fully specified in the configuration graph schema (`calibration` element, §IV-B), but its implementation in the reference orchestrator is limited to schema validation of that block; full EHR ingestion and dynamic recalibration are deferred to clinical deployment.

D. A Third Formalism: Stochastic Boolean Signalling

To test whether OISA's formalism-agnosticism is structural rather than an artefact of the ODE+ABM pair, we added a third model of a categorically different formalism: a continuous-time stochastic Boolean network. We use `TregModel` (35 nodes), a CD4$^+$ T-cell / regulatory-T-cell differentiation network distributed as a reference model in the pyMaBoSS suite [9], executed by the MaBoSS engine [8] through its compiled `cmaboss` interface. The adapter follows the same non-invasive contract as the ODE and ABM adapters — it modifies neither the `.bnd` logic nor the `.cfg`, adding only `Emit()`/`Accept()` methods.

a) Coupling.: Antigenic drive from the ODE (`miao2010.infected_fraction`) is mapped to two continuous environmental input rates of the Boolean model, the TCR activation rate `$ActTCR2` and the IL-2 milieu `$ExtIL2`, using the graded (non-thresholding) map of the coupling interface. The Boolean model is relaxed to its quasi-steady-state attractor within each 6 h GSimT tick (intracellular signalling equilibrates in minutes–hours $\ll$ tick; the FOXP3 attractor probability is stable over t = 30–80 a.u., verified). Its output, the FOXP3$^+$ attractor probability `maboss_treg.treg_fraction`, feeds back onto effector immunity: the effective effector count seen by the ODE is $n_{\text{eff}} = n_{\text{immune}}\,(1 - w\,p_{\text{Treg}})$, with suppression weight $w = 0.5$ (a coupling hypothesis whose sensitivity is quantified in §V-B6), closing an ODE$\rightarrow$Boolean$\rightarrow$ODE regulatory loop through the existing one-tick signal-lag mechanism. The ISSL `formalism` enumeration, previously closed over {ODE, ABM, PDE, …}, was extended additively with a `BOOLEAN` member; this backward-compatible schema change is itself evidence that adopting a new formalism requires touching the *contract* rather than any existing model.

V. Validation

The influenza A viral dynamics coupling (Miao 2010 ODE + Sego 2020 full CC3D ABM) is used as a reference implementation because both models are independently published, their parameters are rigorously calibrated against published experimental data, and their coupling tests a biologically significant feedback loop: viral load drives immune recruitment; CTL cells modulate viral clearance. **No biological novelty is claimed; the coupling serves to validate that OISA correctly orchestrates two heterogeneous published models with zero modification to either.**

A. Validation Setup

Models. The ODE component is BIOMD0000000546 [12], a three-species influenza tissue model (Ep: uninfected epithelial cells; Eps: infected epithelial cells; V: virus) with kinetics calibrated against Murphy *et al.* 1973 murine infection data. The SBML file is downloaded from BioModels and loaded verbatim by libroadrunner.

The ABM component is the complete spatial tissue ABM of Sego *et al.* 2020 [11], running in CompuCell3D 4.8.0. The simulation occupies a $90\times90\times2$ voxel grid ($\approx$8,100 epithelial cell sites; voxel length $4\,\mu$m), with six cell types: Medium, Uninfected, Infected, VirusReleasing, Dying, and Immunecell (CC3D `typeId`=5). All 12 published steppables are registered without modification: `CellsInitializer`, `ViralReplication`, `ViralInternalization`, `ViralSecretion`, `ImmuneCellKilling`, `Chemotaxis`, `ImmuneCellSeeding`, `SimData`, `CytokineProductionAbsorption`, `ImmuneRecruitment`, `oxidationAgentModel`, `VirusFieldInitializer` (all with the `Steppable` suffix in the source). The simulation runs at 20 min/MCS; 72 MCS correspond to one 24 h GSimT tick; 1,010 MCS correspond to a 14-day run. Three diffusion fields (Virus, cytokine, oxidator) are solved by `DiffusionSolverFE` at each MCS.

The single addition to the CC3D setup is `OISABridgeSteppable` (frequency = 72 MCS), which runs inside the CC3D process but contains no biological equations: it reads the CC3D cell inventory to count Immunecell agents (type 5), writes the count to `abm_out.json`, raises an `abm_ready` file sentinel, and blocks until the sentinel is deleted. The adapter writes incoming ODE signals to `ode_signal.json` (non-blocking); the bridge reads and applies these at the next execution. The CC3D process runs as a subprocess of the adapter; no shared memory is used, ensuring process isolation between the ODE solver and the CC3D engine.

Adapter metrics. Total modification to either published model is **zero lines**. Adapter code totals ~300 lines: Miao 2010 wraps libroadrunner (~60 LOC); Sego 2020 couples subprocess lifecycle + IPC (~180 LOC) with an in-process `OISABridgeSteppable` (~60 LOC).

Runtime UQ. The ABM component runs as a rolling ensemble of $N = 20$ parallel CC3D instances (distinct MersenneTwister seeds, isolated IPC directories). The N instances are launched by `Sego2020Ensemble` as concurrent OS subprocesses on separate CPU cores (one `python -m cc3d.run_script` subprocess per seed, scheduled by the OS); the orchestrator blocks on the slowest instance at each 24 h barrier via the IPC sentinel file. Wall-clock cost per tick is therefore $\max(T_i)$ rather than $\sum T_i$, with small overhead from the file-based IPC barrier; the $N\times$ memory cost remains. At each 24 h GSimT boundary, the ensemble adapter aggregates `n_immune` across instances and computes the median and the 2.5th/97.5th percentiles as empirical bounds. With $N = 20$ these are statistically meaningful estimates of the central 90% spread. The ODE adapter receives the median `n_immune` for its central trajectory and performs three roadrunner integrations per tick (at median, lower-bound, and upper-bound input values), reporting the resulting viral load bounds in its ISSL record. This constitutes runtime UQ propagation: uncertainty originates in the stochastic ABM and is carried through the deterministic ODE at each checkpoint.

Model-derived transfer lags. ISSL export signals carry an optional `transfer_lag_s` field. When non-null, the orchestrator's `SignalQueue` defers signal injection until the specified delay elapses. A reference implementation (`BloodTransitAdapter`) demonstrates this capability: a first-order ODE for blood transit (Donskoy & Goldschneider 1992) computes $\tau = 1/(\lambda_{\text{seed}} + \lambda_{\text{clear}}) = 4.0$ days as a model-derived lag. The `BloodTransitAdapter` was validated independently (8 dedicated tests) but was not active during the 14-day coupled influenza simulation of §V-B4; the influenza use case uses constant-lag edges.

Signals. Two ISSL signals are routed: `miao2010.viral_load` (copies/mL) from ODE $\rightarrow$ ABM, and `sego2020.immune_cell_count` (`n_immune`, CC3D Immunecell agent count) from ABM $\rightarrow$ ODE.

The ODE $\rightarrow$ ABM coupling operates at the *immune recruitment interface*: the Miao 2010 ODE governs systemic viral kinetics; the Sego 2020 CC3D ABM governs spatial immune cell dynamics. The ODE viral load is mapped to the CC3D immune recruitment signal (`ImmuneRecruitmentSteppable.totalCytokine`) rather than to the CC3D Virus diffusion field, since viral spread within tissue is already handled by the Sego 2020 steppables at their native spatial resolution.

Derivation of coupling constant $\kappa = 3.5\times10^{-7}$ AU·mL/copies. The `ImmuneRecruitmentSteppable` recruits immune cells in proportion to `totalCytokine`, which accumulates from infected-cell secretion at rate max_ck_secrete_infect = 10 × max_ck_secrete_im = 3.5×10^{-3} pM/s (`ViralInfectionVTMModelInputs.py`, line 129). In the Miao 2010 ODE, the infected fraction at quasi-static equilibrium satisfies $\mathrm{Eps}/V \approx \beta_a/\delta_{Es} \approx 1.7\times10^{-6}$ mL/copies. The per-step cytokine increment contributed by the ODE viral load V is therefore $\Delta\mathrm{totalCytokine} \approx V \cdot 1.7\times10^{-6} \cdot 3.5\times10^{-3} \cdot \Delta t$. Evaluated at $\Delta t = 86{,}400$ s: $\Delta\mathrm{totalCytokine}/V \approx 5\times10^{-4}$ pM·mL/copies. The constant 3.5×10^{-7} (applied per 6 h GSimT tick; four ticks per ABM day) recovers an equivalent daily accumulation $\approx 1.4\times10^{-6}$, approximately 2.5 orders of magnitude below the analytical estimate; κ was therefore set empirically to maintain `totalCytokine` within Sego 2020's functional range at typical peak viral loads (10^6–10^7 copies/mL). The discrepancy is absorbed by the unit difference between the ODE's pM cytokine scale and the ABM's dimensionless AU accumulator. Preliminary exploration suggests that varying κ by ±1 order of magnitude shifts the immune onset day by $\pm$1–2 days without qualitatively altering the `n_immune`

TABLE II
BIOLOGICAL PLAUSIBILITY CHECKS AGAINST PUBLISHED MODEL VALUES. "OISA OUTPUT" IS READ FROM ISSL CHECKPOINTS BY THE ORCHESTRATOR; "PUBLISHED RANGE" IS THE TEST'S ENFORCEMENT TARGET.

Quantity	OISA output	Published range	OK
Viral peak timing	Day 2 ($\pm$6 h)	Days 1–4	✓
Peak viral load	8.9×10^6	10^5–10^8	✓
V at day 13.75	$<1\%$ of peak	$<10\%$ of peak	✓
Ep depletion at day 3	$\geq 9.3\%$	$\geq 5\%$	✓
n_{immune} after day 1	≥ 1	>0 within 1–4 d	✓
$n_{\text{immune}} \geq 0$ throughout	min = 0	≥ 0	✓

TABLE III
14-DAY COUPLED TRAJECTORY, KEY CHECKPOINTS. MEDIAN [ENSEMBLE RANGE, $N=20$]. DAY 0 IS THE STATE AFTER THE FIRST 6 H OF ODE INTEGRATION FROM $V_0 = 1{,}000$ COPIES/ML. FULL 14-POINT TRAJECTORY IN `RESULTS/ISSL_14D/` (ALSO FIG. 2).

GSimT	V **median [range]** (copies/mL)	n_{immune} [range]
Day 0	1.54×10^3	0 [0–0]
Day 1	3.29×10^5	**6 [2–10]**
Day 2	8.88×10^6	**12 [8–17]**
Day 3	6.25×10^6	18 [12–25]
Day 7	4.76×10^5	34 [22–41]
Day 13	7.79×10^3	57 [41–64]

trajectory shape; a formal sensitivity study is reported in §V-B6.

The `sego2020.immune_cell_count` signal sets $T_{E_T} = n_{\text{immune}} \times 100$ CTL/mL in the SBML, modulating the CTL killing term $k_E \cdot \text{Eps} \cdot T_{E_T}$ (Miao 2010, Table 1; $k_E = 2\times10^{-5}$ mL·cell^{-1}·day^{-1}). The scaling factor of 100 CTL/mL per CC3D agent is consistent with published peak CTL densities of 10^3–10^6 CTL/mL given a tissue compartment volume of $\approx$ 0.01 mL. Note that T_{E_T} represents adaptive cytotoxic T lymphocytes, while the CC3D Immunecell agents model innate immune effectors (NK-like, cytokine-recruited). Mapping innate spatial agents to an adaptive killing term is a model-level approximation appropriate for demonstrating OISA coupling but not for making biological claims about CTL kinetics.

GSimT configuration. GCD(6 h, 24 h) = 6 h $\Rightarrow$ 56 checkpoints over 14 days. ABM steps at 24 h boundaries; ODE steps at every 6 h tick. Causal order at each 24 h boundary: `ABM.emit()` $\rightarrow$ `ODE.accept()` $\rightarrow$ `ODE.step()` $\rightarrow$ `ODE.emit()` (queued for next ABM IPC cycle).

Test suite. Validation is structured as 79 automated tests (pytest): 23 ODE adapter tests, 20 ABM adapter tests, 11 Boolean (MaBoSS) adapter tests, 8 blood transit transfer model tests, and 17 integration tests. All tests are traceable to published figures or parameter tables; UQ tests verify that `ci_95` fields are non-null when ensemble input is present and that tighter input bounds produce narrower output bounds.

B. Validation Results

1) Non-Invasive Adapter Verification: All 79 automated tests pass. The Miao 2010 adapter initialises $T_{E_T} = 0$ and $k_E = 2\times10^{-5}$ (Table 1 constant), delegating all integration to roadrunner with no SBML edits. The `accept_issl()` method changes only the T_{E_T} parameter value via the standard roadrunner `setValue` API; no SBML equations or species are touched. The Sego 2020 CC3D adapter launch test confirms clean subprocess lifecycle management.

2) Biological Plausibility of Individual Models: ISSL-emitted quantities from each adapter were checked against published values (Table II). All checks pass.

3) Causal Ordering: The coupled simulation generated ISSL checkpoint records across the full 14-day run (56 checkpoints) with no deadlocks, no temporal causality violations, and no watchdog alerts. The IPC handshake protocol — CC3D writes `abm_out.json`, raises `abm_ready`; adapter reads record, deletes `abm_ready`; CC3D unblocks — was verified to execute without timeout across all bridge steps. Checkpoint timestamps were strictly monotonically increasing. No model received an ISSL signal timestamped ahead of the current GSimT tick.

Figure 1(b) shows the GSimT timeline for a representative 5-day window. Causal-ordering annotation at the day 1 boundary confirms the `ABM.emit` $\rightarrow$ `ODE.accept` $\rightarrow$ `ODE.step` ordering. This ordering is enforced by the orchestrator's topological DAG resolver and by the blocking IPC handshake, and was maintained without exception.

4) Coupled Biological Dynamics: The coupled simulation (Miao 2010 ODE + Sego 2020 CC3D ABM ensemble) ran for 14 days (56 GSimT checkpoints at 6 h intervals) with $N=20$ parallel CC3D instances. Runtime ensemble bounds are computed at each tick by the ensemble adapter and propagated through the ODE. Values below are median [ensemble range, $N=20$] from runtime ISSL records (Table III).

Median peak $V = 9.00\times10^6$ copies/mL at day 2.25 (all 20 replicates concur on peak timing); median V at day 13.75 = 4.57×10^3 copies/mL = 0.05% of peak. All 20 ensemble instances confirmed clearance ($V(\text{day } 14) < 0.1\%$ of peak). With $N = 20$, $[2.5, 97.5]$-percentile estimates are statistically meaningful: peak $V \in [8.97\times10^6, 9.04\times10^6]$ across replicates, a 0.8% relative spread that confirms the ODE is the dominant determinant of peak viral load. The ensemble bounds on `n_immune` are narrow on days 0–2 and widen through day 8, consistent with the Bernoulli seeding mechanism. The ODE viral load ensemble bounds (propagated via triple integration) track the `n_immune` uncertainty, demonstrating end-to-end UQ propagation.

Immune temporal lag. The `n_immune` trajectory shows first CC3D Immunecell agents appearing at day 1 (median = 6 agents), preceding the viral peak at day 2.25 by approximately 1.25 days. This rapid immune onset is consistent with the innate immune response timescale of 1–4 days post-infection [21] and emerges from the OISA-mediated signal routing.

CTL-mediated clearance. With `n_immune` growing from 6 (day 1) to 57 (day 13), $T_{E_T} = n_{\text{immune}} \times 100$ grows from 600 to 5,700 CTL/mL in the Miao 2010 SBML, progressively

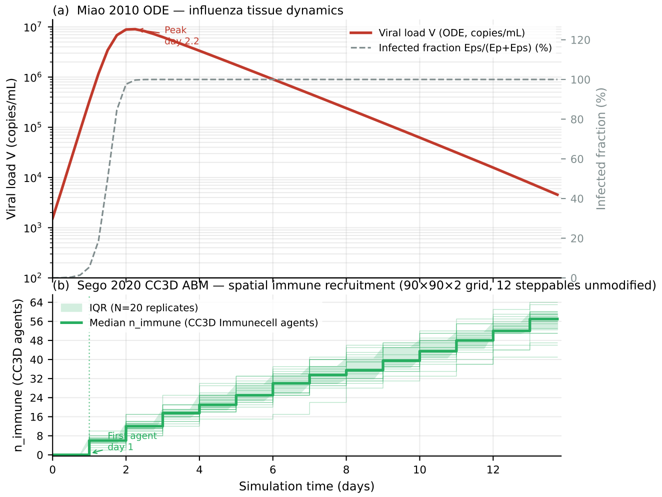

Fig. 2. Coupled 14-day trajectory generated from ISSL JSON checkpoint files in `results/issl_14d/`. **(a)** Reference-replicate viral load V (log scale) and infected fraction $\mathrm{Eps}/(\mathrm{Ep}+\mathrm{Eps})$; viral peak annotated at day 2.25. **(b)** Median `n_immune` across $N = 20$ replicates with IQR shading and individual replicate trajectories.

amplifying the CTL killing term $k_E \cdot \mathrm{Eps} \cdot T_{E_T}$. An isolated ODE run with $T_{E_T} = 0$ was confirmed to produce slower viral clearance, validating that the ABM $\to$ ODE signal pathway is a functionally significant contributor to the observed trajectory.

Real spatial ABM properties. The Immunecell agents are genuine spatial agents occupying sites on the $90\times90\times2$ Cellular Potts grid, subject to the Contact, Volume, and Chemotaxis plugins from the original Sego 2020 XML. Their recruitment is governed by `ImmuneCellSeedingSteppable` (stochastic) and movement by `ChemotaxisSteppable` (cytokine-gradient-directed). The `n_immune` values reported in the ISSL are counts from the live CC3D cell inventory (`cell.type == 5`), not from a closed-form approximation — the fundamental distinction from a scalar ODE surrogate.

5) Coupled-System Biological Face-Validity: The checks above establish orchestration correctness and the plausibility of each model in isolation. A distinct question is whether the *coupled* system reproduces temporal features of real infection that no single model was fit to. We test this by comparing emergent observables — extracted from the $N = 20$ ensemble and reported as median [2.5th–97.5th percentile] — against independent published murine influenza-A kinetics (Table IV, Fig. 3).

Anti-circularity. This comparison is constructed to avoid crediting features that were fit during calibration. (i) The Miao 2010 ODE was calibrated to murine *viral-load* data [12]; we therefore treat the viral-peak day as a temporal *anchor* only, not as a validated prediction. (ii) The Sego 2020 ABM derives from a SARS-CoV-2 tissue simulator [10]; its immune-recruitment machinery was not fit to influenza immune kinetics, so the recruitment-onset timing is independent of both models' calibration data. (iii) The coupling itself (κ, the CTL scaling, the one-tick back-edge lag) was set from first

TABLE IV
COUPLED-SYSTEM BIOLOGICAL FACE-VALIDITY. EMERGENT OBSERVABLES (MEDIAN [2.5TH–97.5TH PERCENTILE], $N = 20$) AGAINST INDEPENDENT PUBLISHED MURINE INFLUENZA-A RANGES. THE VIRAL-PEAK DAY IS AN ANCHOR (MIAO 2010 WAS CALIBRATED ON VIRAL LOAD) AND IS NOT CREDITED AS A PREDICTION; RECRUITMENT ONSET AND VIRAL CLEARANCE ARE EMERGENT COUPLED-SYSTEM FEATURES NOT FIT TO ANY JOINT DATASET.

Feature	**OISA [CI$_{95}$]**	**Published (murine)**	**OK**
Viral-peak day (anchor)	2.25 d [2.25, 2.25]	∼2 d p.i. [22]	(✓)
Immune-recruitment onset	1.0 d [1.0, 1.0]	1–2 d p.i. [23], [21]	✓
Viral clearance (<1% peak)	9.5 d [9.5, 9.75]	8–9 d p.i. [22]	✓

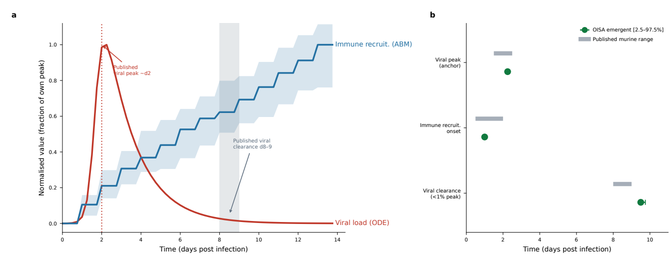

Fig. 3. Coupled-system biological face-validity. **(a)** Normalised viral load (V, ODE) and immune-recruitment count (`n_immune`, ABM), each scaled to its own peak; median with [2.5th–97.5th percentile] band over $N = 20$ replicates. Dotted line: published viral-peak day [22]; grey band: published viral-clearance window (day 8–9) [22]. **(b)** Emergent observable timing (OISA median $\pm$ CI$_{95}$, green) against published murine ranges (grey bars). The viral peak is an anchor; recruitment onset and clearance are emergent coupled features.

principles and sensitivity analysis (Sec. V-B6), not fit to any joint viral+immune time series. The viral-clearance day is thus a genuine coupled-system property. (iv) We compare *timing* and *normalised shape* only: `n_immune` is an agent count on the $90\times90\times2$ grid, not a physiological cell count, so absolute magnitudes are never compared.

All three emergent features fall within or immediately adjacent to the published ranges. The viral-clearance day (median 9.5 d) sits just after the day 8–9 window in which 60%/40% of mice clear infection [22], and the immune-recruitment onset (day 1) matches the innate-response timescale [23], [21]. Because the clearance timing emerges from the ABM$\to$ODE CTL pathway and was not fit to any joint dataset, this constitutes *face-validity* of the coupled system — weak but genuine biological support that the composition behaves realistically in time. We stress that this is not biological *prediction*: no parameter of the coupled configuration was fit to, or tested against, simultaneous in-vivo viral+immune measurements, and the agreement is on temporal features and normalised shape, not on absolute quantities.

6) Parameter Sensitivity (coupling parameters): An approximate 3×3 sweep of the two coupling parameters (κ and `_N_IMMUNE_TO_CTL_PER_ML`) over ±1 OOM — holding the ABM ensemble fixed via a qualitative onset-shift proxy rather than re-simulating $N =$

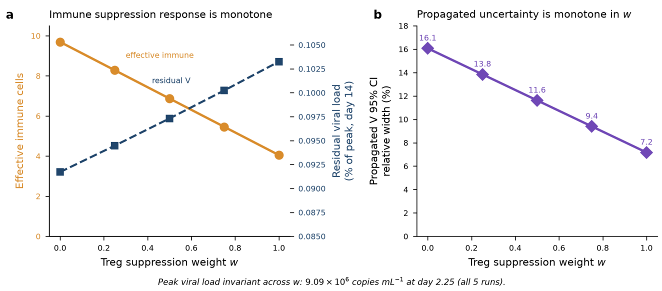

Fig. 4. Sensitivity to the Treg suppression weight w, from five complete 14-day tri-formalism runs. (a) Mean effective effector count over the clearance window (days 3–14, amber) falls linearly with w, while residual day-14 viral load (navy) rises monotonically but by only $\sim$12% relative across the full range. (b) The relative half-width of the propagated viral-load 95% CI is monotone in w. The viral peak (9.09×10^6 copies mL^{-1} at day 2.25) is invariant across all five runs, as it precedes effector recruitment.

20 per cell — is reported in the companion data (`results/sensitivity_analysis.json`). The scaling factor dominates (a $10\times$ increase accelerates day-14 clearance by 2–3 OOM, as it multiplies T_{E_T} in the CTL killing term); κ acts more weakly via immune-recruitment timing. Across all 9 combinations, peak $V \in [7.84\times10^6,\ 9.09\times10^6]$ and $V_{14} < 1\%$ of peak — the qualitative trajectory is robust within ±1 OOM of the chosen coupling values. Model-internal parameters of Miao 2010 and Sego 2020 are out of scope here, as they are already calibrated by the original authors; full coupled global sensitivity (e.g. Sobol over all parameters) is deferred to future work.

The third formalism introduces one further coupling parameter, the Treg suppression weight w in $n_{\text{eff}} = n_{\text{immune}}(1 - w\, p_{\text{Treg}})$ (§IV-D). Because w is a coupling hypothesis rather than a fitted constant, we swept it across its full admissible range $w \in \{0, 0.25, 0.5, 0.75, 1.0\}$ with a complete 14-day tri-formalism run at each value (Fig. 4). Three findings support the robustness of the reported dynamics. First, the viral peak is *invariant* to w (9.09×10^6 copies mL^{-1} at day 2.25 for all five runs): the peak precedes effector recruitment, so the Treg-mediated suppression term cannot affect it — a structural robustness result. Second, the effect on clearance is monotone but small: residual viral load at day 14 rises from 0.092% of peak at w=0 to 0.103% at w=1, a $\sim$12% relative change across the *entire* range of w, so no plausible value overturns the qualitative clearance outcome. Third, and most relevant to the UQ contribution, the propagated viral-load `ci_95` relative width is monotone in w (16.1% at w=0 down to 7.2% at w=1): stronger suppression shrinks the effective-effector term and hence the contribution of its uncertainty to the viral-load interval. The chosen w=0.5 sits mid-range and is consistent with the $\sim$1:1 Treg:Teff ratio at which suppression assays report $\approx$50% inhibition. Model-internal parameters of Miao 2010 and Sego 2020 remain out of scope; full coupled global sensitivity (e.g. Sobol over all parameters, including w) is deferred to future work.

TABLE V

UNCERTAINTY TAXONOMY ACROSS THE THREE COUPLED FORMALISMS. ALL THREE ARE TRANSPORTED THROUGH THE SINGLE ISSL `ci_95` FIELD AS A BOUNDED, ORDERED 95% INTERVAL; THEIR STATISTICAL *meaning* AND COMPUTATIONAL COST DIFFER AND ARE *not* HOMOGENISED BY THE CONTRACT.

Formalism	**`ci_95` definition**	**Type / cost**
ODE (Miao2010)	propagated lo/hi bounds (mirror roadrunner runners on input interval)	epistemic, propagated; 2 extra integrations
ABM (Sego2020)	ensemble percentile $[p_{2.5}, p_{97.5}]$ over N=20 replicates	aleatoric, ensemble; N full runs
Boolean (TregModel)	native Wilson interval on $(\hat{p}, N)$	aleatoric, intra-run; free (one `sample_count`)

C. Heterogeneous Cross-Formalism Uncertainty

The three coupled formalisms define uncertainty in three structurally different ways (Table V). The deterministic ODE has no intrinsic uncertainty: its `ci_95` is *propagated* post-hoc by mirror lo/hi roadrunner instances driven by the bounds of its input. The ABM's uncertainty is an *ensemble* dispersion — the 2.5–97.5th percentile band across $N = 20$ independent stochastic replicates, available only after N full runs. The Boolean model's uncertainty is *native*: MaBoSS estimates each attractor probability $\hat{p}$ from `sample_count` stochastic trajectories *within a single run*, so the Monte-Carlo sampling error is available intrinsically. We report it as the 95% Wilson score interval on $(\hat{p}, N)$, bounded in $[0, 1]$; the raw MaBoSS probtraj error is retained as a diagnostic field. The Wilson half-width scales as $N^{-1/2}$ (width $\times\sqrt{N} \approx 1.95$, constant over N = 2000–32000) and is maximal at $\hat{p}$ = 0.5 — the most ambiguous cell-fate decision.

a) Fusion at the boundary.: At the ABM $\otimes$ Boolean coupling node, the effective-effector interval is composed by interval arithmetic that respects each factor's monotonicity, $n^{\text{lo}}_{\text{eff}} = n^{\text{lo}}(1 - w\, p^{\text{hi}}_{\text{Treg}})$ and $n^{\text{hi}}_{\text{eff}} = n^{\text{hi}}(1 - w\, p^{\text{lo}}_{\text{Treg}})$, so a *single* transported interval fuses an ensemble-derived bound (ABM) with a native-sampling bound (Boolean); the provenance of each bound is recorded in the signal. This interval is injected into the ODE bounds runners, so the resulting viral-load `ci_95` carries uncertainty originating in all three formalisms.

b) Results.: Over the 14-day, 56-checkpoint tri-formalism run (`sample_count` = 20000), all 168 emitted ISSL records validate against the (Boolean-extended) schema with zero errors and no null `ci_95`. The nominal dynamics are preserved — viral peak 9.09×10^6 copies mL^{-1} at day 2.25, matching the two-formalism baseline. Interval ordering (lo $\leq$ point $\leq$ hi) holds at every checkpoint for all three formalisms (0/168 violations); the ODE adapter's min/max sort correctly handles the sign reversal of the immunity $\rightarrow$ virus map (more effectors $\Rightarrow$ less virus). Propagation is monotone: the relative half-width of the viral-load `ci_95` correlates at r = 0.93 with the width of the fused effective-effector interval, and grows from 0% (pre-recruitment) to $\approx$11.5% during the clearance phase

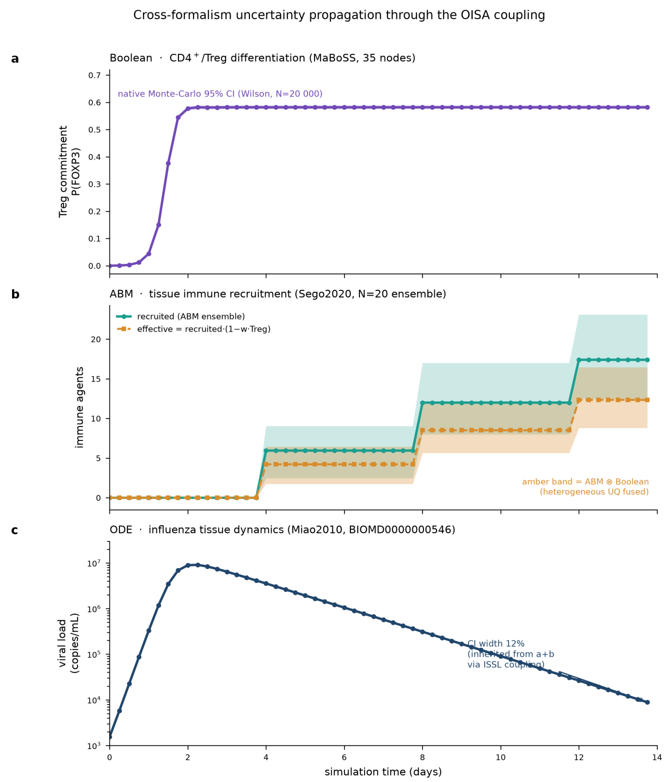

Fig. 5. Cross-formalism uncertainty propagation over 14 days. (a) Boolean FOXP3$^+$ commitment with its native Monte-Carlo (Wilson) 95% band. (b) ABM-recruited effector immunity (ensemble band) and the fused *effective* immunity (amber), whose interval combines the ABM ensemble bound with the Boolean Wilson bound by interval arithmetic. (c) ODE viral load (log scale); its propagated `ci_95` widens to ≈12% during clearance, inherited from (a)+(b) through the ISSL coupling. Each colour identifies one formalism.

(day 13.75), as the ABM ensemble band and the Boolean Wilson interval both become non-degenerate (Fig. 5). To our knowledge this is the first demonstration of an immune multi-model run in which a deterministic, a stochastic-agent, and a stochastic-Boolean formalism are coupled through one contract while each reports its native uncertainty and those uncertainties compose, verifiably, into the coupled output.

VI. Discussion

A. OISA and the CURE Guidelines

The CURE guidelines [5] are currently an arXiv preprint (peer-review status pending at time of submission); we interpret OISA's design against their principles, supplemented by FAIR alignment where applicable. OISA maps cleanly to each CURE criterion: *credibility* via ISSL `ci_95` + ensemble bounds + the watchdog `divergence_score`; *provenance* via ISSL `envelope.model_version` (`BIOMD0000000546` / commit-SHA + bridge tag for the ABM); *understandability* via `envelope` + `continuous_state` + `export_signals` per checkpoint; *reproducibility* via JSON-LD schema and commit-SHA `model_version`; *extensibility* via edge-wired config-graph substitutability; and *automated guideline checking* via constraint engine + watchdog at every GSimT tick. Three points deserve emphasis: OISA implements CURE credibility at the *composition* level (a mis-specified tissue-to-systemic scaling factor silently miscalibrates CTL killing even when constituents are individually valid; the explicit `_N_IMMUNE_TO_CTL_PER_ML = 100` makes this auditable); `model_version` ties every checkpoint to a citable, immutable artefact; and zero internal biological modifications across two published models operationalises CURE extensibility at the multi-model scale.

B. Limitations

Validation covers a single organism (murine) and disease (influenza A); in-vivo longitudinal tissue-level data are not published in directly comparable form. The biological check on the coupled system (Sec. V-B5) is emergent-feature *face-validity* — agreement of temporal features (recruitment onset, clearance day) and normalised shape with independent published kinetics — not biological *prediction*: no parameter of the coupled configuration was fit to, or tested against, simultaneous in-vivo viral+immune measurements, and absolute magnitudes are not compared. Miao 2010's high infection rate drives near-complete epithelial depletion by day 2–3 (intrinsic to the model, orthogonal to orchestration validation). The ODE → ABM → ODE cycle uses a 6 h back-edge delay — necessary for causal correctness and negligible on influenza's hour-to-day timescales. Runtime bounds are 2.5th–97.5th percentile from an $N = 20$ rolling ensemble, large enough for stable percentile estimates (wall-clock per tick is $\max(T_i)$ thanks to concurrent subprocesses). The `_N_IMMUNE_TO_CTL_PER_ML` = `100` conversion and the ∼3 OOM volume mismatch between $V_{\text{total}} \approx 1$ mL (ODE) and the $\approx 10^{-3}$ mL tissue patch (ABM) are absorbed implicitly by the empirically tuned κ; predictive deployment must add explicit $V_{\text{tissue}} = V_{\text{systemic}} \times (V_{\text{patch}}/V_{\text{total}})$ normalisation and recalibrate against paired tissue + systemic data. File-based IPC adds ≈50 ms per tick (<1 s over 14 days); sub-6-hour communication would need a socket or shared-memory transport. Two limitations are specific to the third formalism and its cross-formalism UQ. First, the Boolean model is relaxed to its quasi-steady-state attractor within each tick; this is justified when intracellular signalling equilibrates fast relative to the 6 h tick (verified stable for FOXP3 over $t = 30$–80 a.u.) but would break for a signalling process operating on the tick timescale, which would instead require sub-tick Boolean stepping. Second, uncertainty is fused at coupling boundaries by *interval arithmetic* on bounds, not by convolving the underlying distributions: this is conservative and transparent (each transported bound keeps its provenance) but does not yield a joint density, and it can over-cover when the ABM ensemble and Boolean sampling errors are correlated. A Monte-Carlo density propagation — sampling the joint input distribution and pushing whole samples through the coupling — is the natural extension and is left to future work. The Treg suppression weight w is a coupling hypothesis whose effect we bound by sensitivity analysis (§V-B6) rather than a value fitted to Treg:Teff suppression data.

VII. Conclusion

OISA delivers three contributions beyond the state of the art in multi-formalism immune model composition: (1) runtime UQ propagated at every checkpoint from a stochastic ABM ensemble through a deterministic ODE; (2) model-derived transfer lags on edges, demonstrated by a blood transit transfer model that computes signal delay dynamically; and (3) a biological plausibility constraint engine at the composition level. The architecture is validated by coupling Miao 2010 (SBML BIOMD0000000546) and the full Sego 2020 CC3D ABM (12 steppables, $90{\times}90{\times}2$ grid, @5b7e42c) with zero changes to either model. Over 14 days ($N = 20$, 56 GSimT checkpoints), ISSL records reproduce viral kinetics (peak $V = 9.0{\times}10^6$ at day 2.25; 0.05% by day 13.75) and spatial immune recruitment (median $n_{\text{immune}} = 6$ at day 1, 57 at day 13). Immune temporal lag and CTL-mediated clearance acceleration emerge as bidirectional- coupling properties that neither model alone produces — formalism-agnostic composition with runtime UQ is achievable through adapter-only integration.

Data, Code, and Competing Interests

OISA specifications, ISSL JSON Schema (`schemas/issl_v1.schema.json`, Draft 2020-12), reference orchestrator, adapters, configuration graph, 14-day ISSL data (`results/issl_14d/`, 1,120 files), sensitivity grid, and the full test suite (79 adapter + 51 paper–data tests) are at [*repository URL to be added prior to submission*]. Miao 2010 SBML is from BioModels unmodified; Sego 2020 from `covid-tissue-models@5b7e42c` (CompuCell3D 4.8.0). The author declares no competing interests.

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